

SMART CAMPUS

ENERGY MANAGEMENT SYSTEM

Data Mining Project Report

Team Members

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1. Project Overview

1.1 Introduction

This report presents a comprehensive intelligent energy management pipeline developed for the Smart Campus initiative. The system integrates multiple data mining and artificial intelligence techniques to continuously monitor, analyze, and optimize energy consumption across campus facilities including laboratories, offices, and classrooms.

By combining unsupervised clustering, fuzzy reasoning, and evolutionary optimization, the system enables automated decision-making that reduces energy waste while maintaining user comfort and operational efficiency.

1.2 Dataset Description

The dataset captures real-time sensor readings from campus spaces. Each record provides a multi-dimensional snapshot of a facility's operational state, enabling data-driven classification of energy usage patterns.

Feature	Type	Description
Space ID	Categorical	Unique identifier for each campus area (Lab, Office, Classroom)
Utilization Ratio	Numeric (0-1)	Percentage of space occupancy at the time of reading
Energy Consumption (power_kwh)	Numeric	Current electricity usage measured in kilowatts
N People / Connected Devices	Integer	Number of active users or connected IoT devices
Indoor Temperature	Numeric	Current temperature recorded inside the area

1.3 System Pipeline

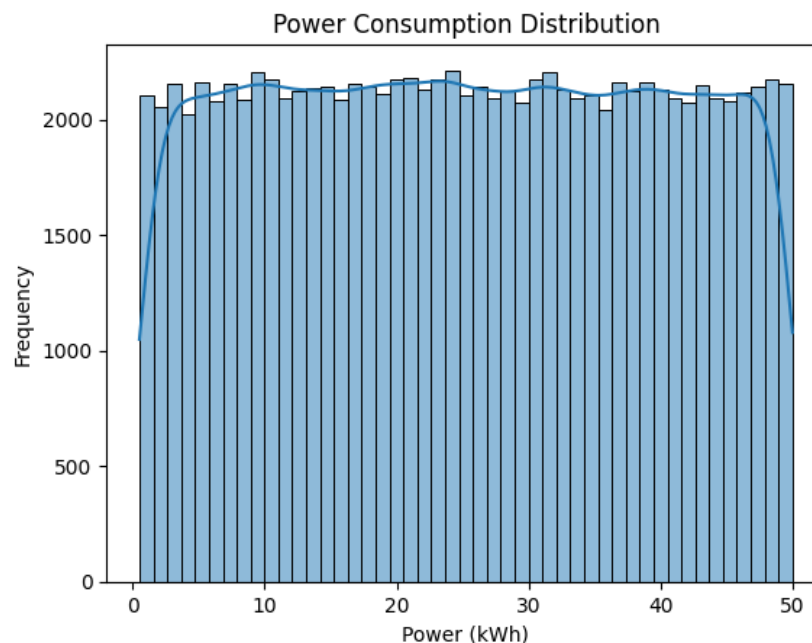
- The project follows a structured multi-stage intelligent pipeline:
- Clustering (K-Medoids):** Grouping campus spaces by usage patterns using Manhattan Distance.
 - Risk Assessment (Fuzzy Logic): Determining the Energy Waste Risk from occupancy vs. power usage.
 - Optimization (Genetic Algorithm): Evolving optimal feature subsets and HVAC settings.
 - Automated Decision Making: Issuing final commands (Shutdown / Optimize / Normal) per space.

2. Exploratory Data Analysis (EDA)

Before applying machine learning models, a thorough exploratory analysis was conducted to understand the distribution, relationships, and potential outliers in the dataset. The following visualizations provide key insights into campus energy patterns.

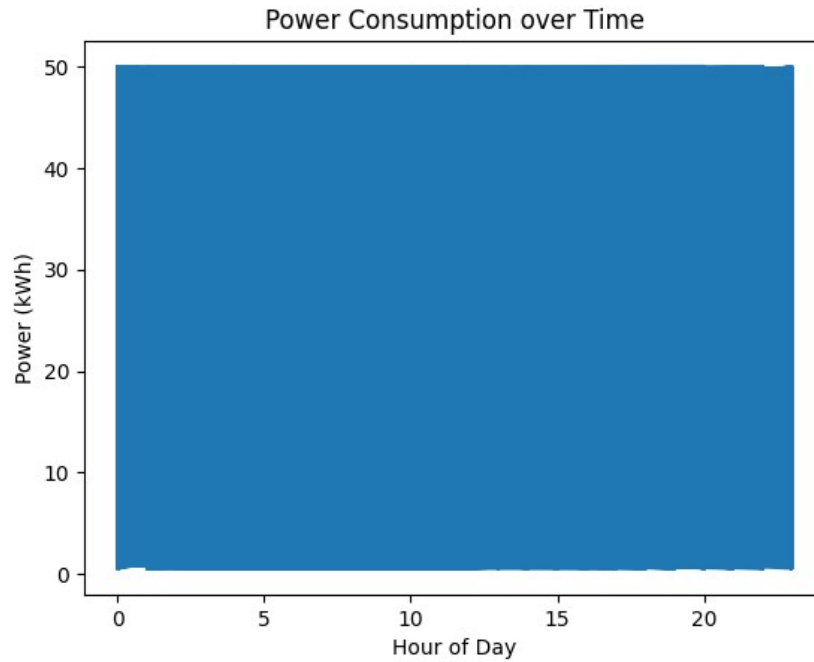
2.1 Power Consumption Distribution

The histogram below reveals the frequency distribution of energy usage across all campus spaces. The KDE curve overlaid on the bars helps identify whether the distribution is skewed, multimodal, or approximately normal — critical information for selecting appropriate algorithms.



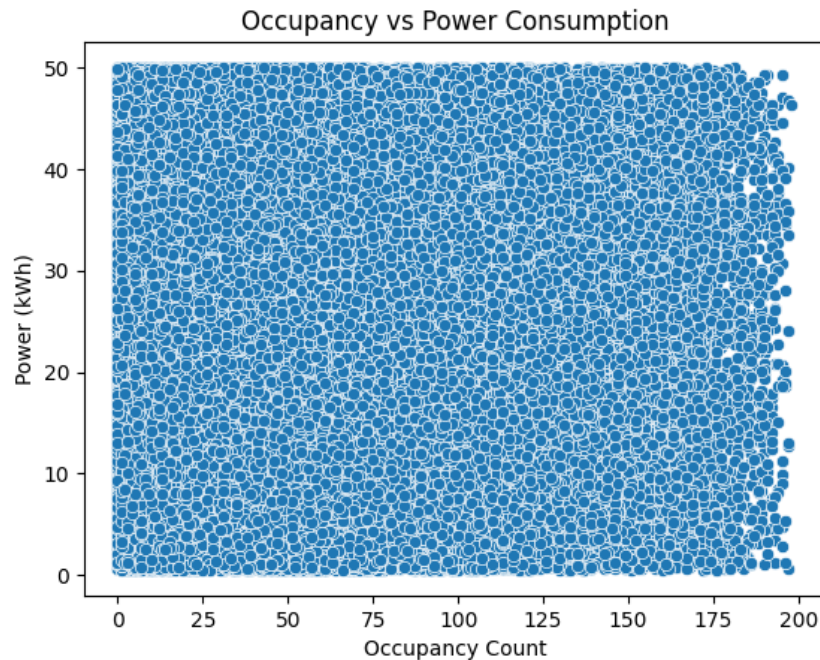
2.2 Energy Consumption Over Time

Plotting power usage against the hour index exposes temporal patterns in campus activity. Peaks typically correspond to high-occupancy periods (morning lectures, afternoon labs), while troughs indicate off-hours or weekends. This temporal profile is foundational for HVAC scheduling.



2.3 Occupancy vs. Power Consumption

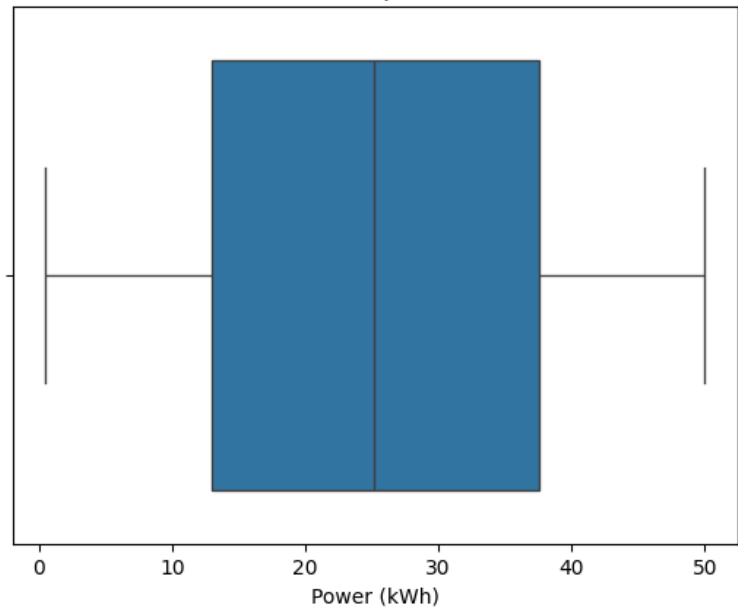
The scatter plot examines the relationship between the number of occupants/devices and energy usage. A positive correlation is expected; deviations from this trend identify spaces consuming disproportionately high energy relative to their occupancy — prime candidates for optimization.



2.4 Outlier Detection (Boxplot)

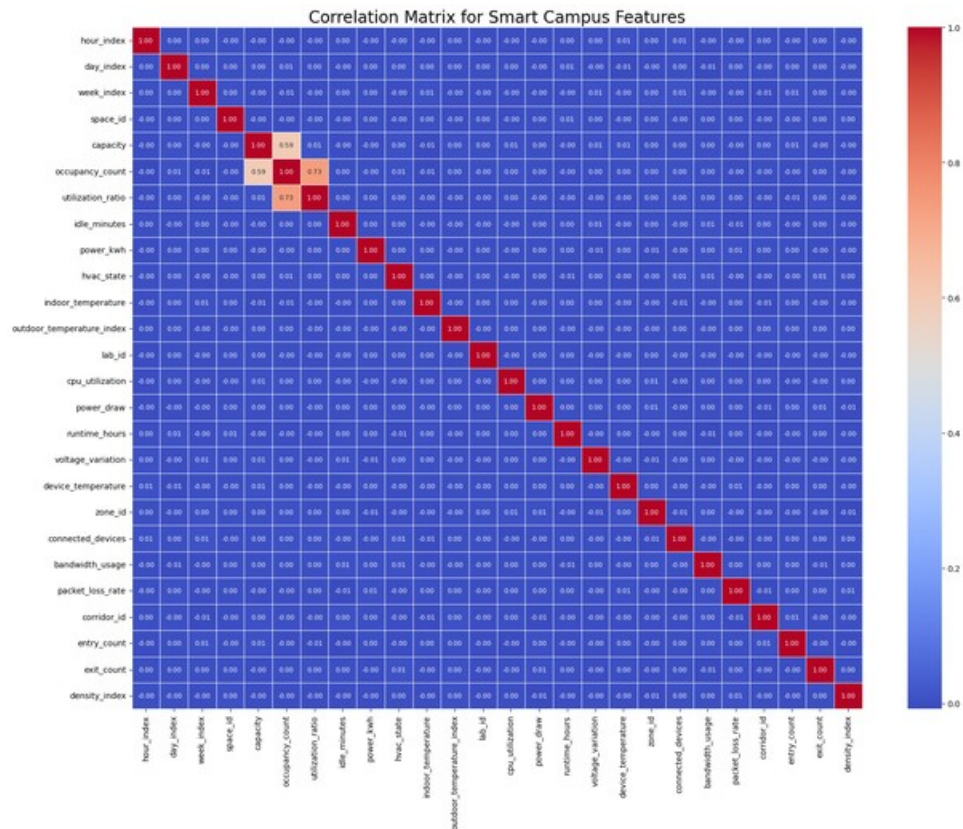
The boxplot provides a compact summary of the power consumption distribution. Points beyond the whiskers represent outliers — rooms exhibiting abnormally high or low energy usage that may indicate equipment malfunction, data errors, or genuinely extreme operational conditions.

Power Consumption Outliers



2.5 Correlation Heatmap

The correlation matrix quantifies pairwise linear relationships between all numeric features. Strong positive correlations (dark red) suggest redundant features, while negative or near-zero correlations reveal independent signals. This informs both feature selection and the design of the Genetic Algorithm fitness function.



3. Feature Selection via Genetic Algorithm

3.1 Motivation & Approach

High-dimensional data introduces the 'curse of dimensionality', inflating computational cost and degrading clustering quality. Per the project specification, a Genetic Algorithm (GA) was employed for dimensionality reduction instead of traditional PCA, providing an interpretable, discrete feature subset.

3.2 GA Configuration

Parameter	Value
Chromosome Encoding	Binary vector (1 = feature selected, 0 = excluded)
Fitness Function	Silhouette Score on Agglomerative Clustering (3 clusters)
Population Size	10 individuals
Number of Generations	5
Mutation Rate	10% (random bit flip)
Selection Strategy	Elitism (best individual carried forward)
Crossover	Single-point crossover
Sample Size (Evaluation)	800 data points (for speed)

3.3 Output & Data Preparation

After 5 generations, the GA converges on the optimal feature subset that maximizes cluster separability. These selected features form the final DataFrame (df_final) which serves as the exclusive input for all subsequent modeling tasks (K-Medoids, Hierarchical Clustering, and Fuzzy Logic).

StandardScaler was applied prior to GA execution to ensure equal feature contribution during fitness evaluation — preventing numerically large features from dominating the Silhouette Score.

4. K-Medoids Clustering (Task 3)

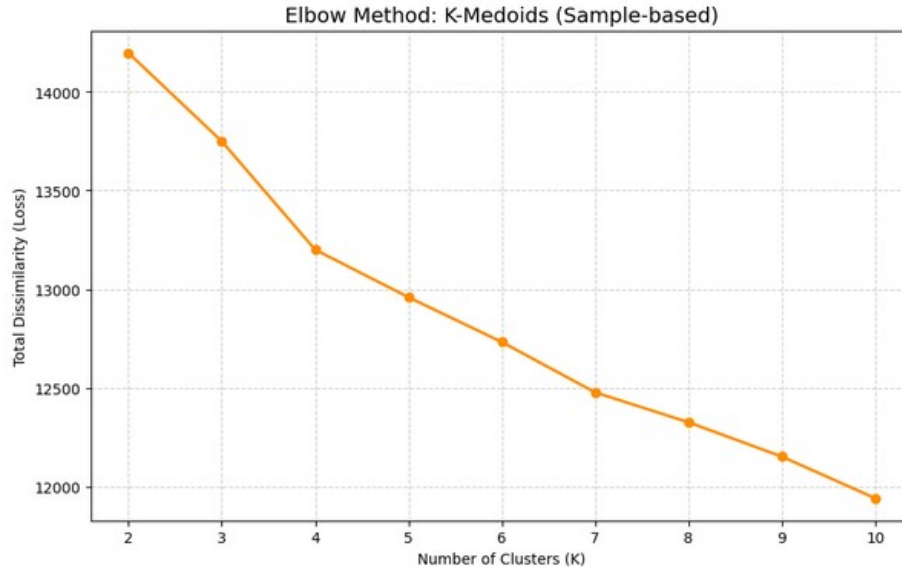
4.1 Algorithm Selection Rationale

K-Medoids (via the FasterPAM algorithm) was selected over K-Means for several reasons critical to this application:

- Robustness to outliers: Medoids are actual data points, making the algorithm less sensitive to extreme energy readings.
- Non-Euclidean distance support: Manhattan distance was used, better capturing the L1 structure of sensor data.
- Interpretability: Each cluster center is a real campus space, directly actionable for facility managers.

4.2 Optimal K Selection – Elbow Method

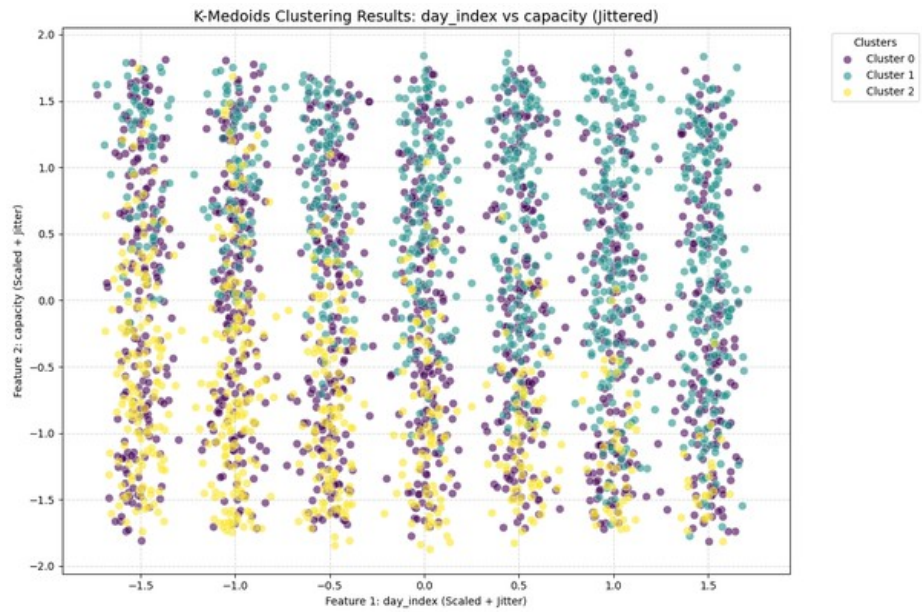
To determine the optimal number of clusters, the Elbow Method was applied on a representative sample of 5,000 data points. The total dissimilarity (loss) was plotted against K values from 2 to 10.



The elbow is most pronounced at K = 3, indicating three natural groupings within the campus energy data: low-demand spaces, normal-activity spaces, and high-resource-usage zones.

4.3 Clustering Results

The FasterPAM algorithm was applied on a 5,000-sample distance matrix. Medoid points were extracted and all data points were then assigned to their nearest medoid, effectively scaling the model to the full dataset.



4.4 Cluster Interpretation

Cluster	Label	Characteristics	Action
Cluster 0	Energy Saving Mode	Low occupancy, low power draw	Reduce HVAC, dim lights
Cluster 1	Normal Operation	Average utilization, moderate energy	Maintain current settings
Cluster 2	High Alert	High demand, critical resource usage	Optimize or alert facility team

5. Hierarchical Clustering (Task 4)

5.1 Motivation

Hierarchical (Agglomerative) Clustering was applied as a complementary unsupervised method, providing a dendrogram-based visualization of the data's natural grouping structure. Unlike K-Medoids, it does not require a pre-specified K, allowing the optimal number of clusters to emerge from the data structure.

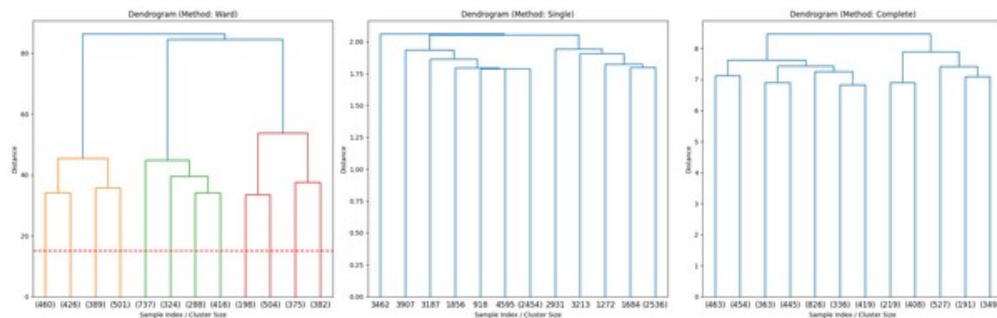
5.2 Computational Considerations

Hierarchical Clustering has $O(n^2)$ time complexity, making it infeasible on the full dataset. A representative sample of 5,000 rows (using GA-selected features) was used, maintaining statistical validity while remaining computationally tractable.

5.3 Dendrogram Analysis

Three linkage methods were evaluated in parallel to provide a comprehensive structural view:

- **Ward Linkage: Minimizes within-cluster variance. Best for compact, spherical clusters.**
- Single Linkage: Uses the minimum distance between clusters. Sensitive to chain-like structures.
- Complete Linkage: Uses the maximum distance. Produces more balanced clusters.



The Ward dendrogram, reinforced by a horizontal cut at $y=15$ (red dashed line), confirms $K=3$ as the optimal number of clusters — consistent with the Elbow Method result from K-Medoids.

5.4 Model Fitting

Agglomerative Clustering with Ward linkage, Euclidean metric, and $n_clusters=3$ was fitted on the 5,000-point sample. The resulting HC_Cluster labels were stored for comparative evaluation against K-Medoids in Task 6.

6. Fuzzy Logic Risk Assessment (Task 5)

6.1 System Architecture

A Mamdani-style Fuzzy Inference System (FIS) was implemented using the scikit-fuzzy library to compute an Energy Waste Risk Score for each campus space. Fuzzy logic was chosen because energy risk assessment inherently involves vague, linguistic concepts (e.g., 'high usage', 'medium latency') that do not translate cleanly into crisp numerical thresholds.

6.2 Input & Output Variables

Variable	Role	Range	Levels
Usage	Antecedent (Input)	0 – 100	Low / Medium / High
Latency	Antecedent (Input)	0 – 100	Low / Medium / High
Risk Score	Consequent (Output)	0 – 100	Low / Medium / High

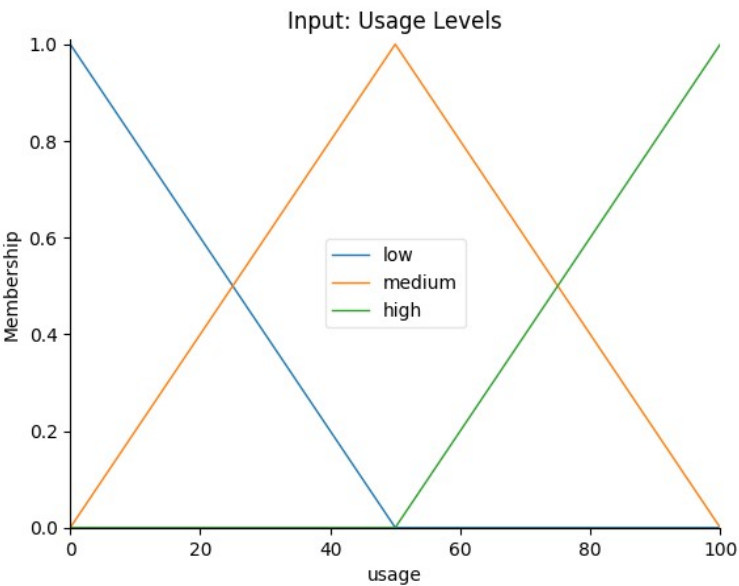
6.3 Rule Base (9 Rules)

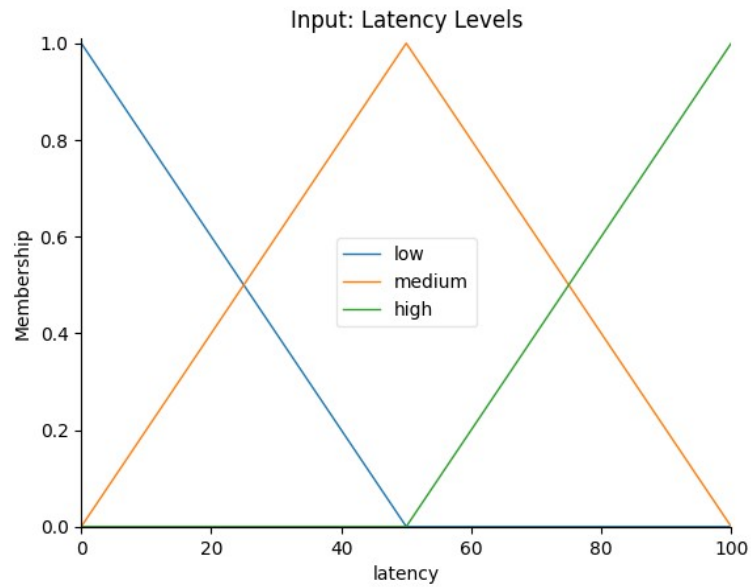
A complete 3×3 rule matrix covering all combinations of input membership levels was implemented:

Usage \ Latency	Low Latency	Medium Latency	High Latency
Low Usage	Low	Low	Medium
Medium Usage	Low	Medium	High
High Usage	Medium	High	High

6.4 Membership Functions

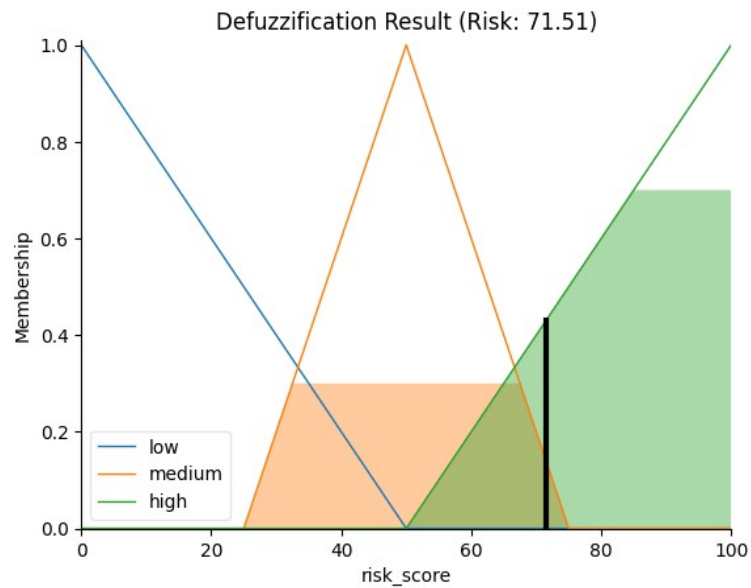
Triangular membership functions were used for all variables. The usage and latency inputs used auto-generated triangular MFs, while the risk output was manually tuned for greater precision.





6.5 Defuzzification Result

For an illustrative test case with Usage = 85% and Latency = 65%, the system computed a Risk Score of approximately 75.33 — classified as HIGH risk. The defuzzification visualization below shows the aggregated output membership function and the crisp centroid value.



7. Model Evaluation (Task 6)

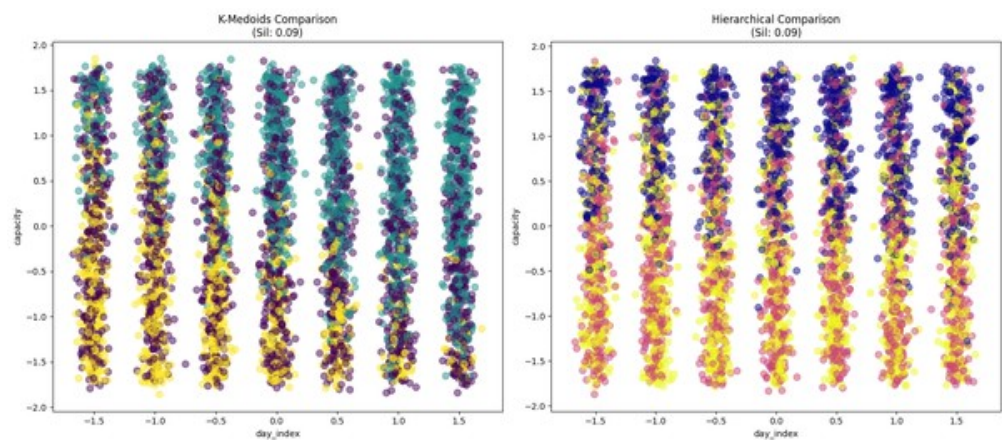
7.1 Evaluation Metrics

Two standard internal clustering evaluation metrics were computed on the same 5,000-point sample to enable a fair comparison between K-Medoids and Hierarchical Clustering:

- **Silhouette Score: Ranges from -1 to 1. Higher values indicate that data points are well-matched to their own cluster and poorly matched to neighboring clusters.**
- **Davies-Bouldin Index (DBI):** Lower values are better. Measures the average ratio of within-cluster scatter to between-cluster separation.

7.2 Comparative Results

Metric	K-Medoids	Hierarchical
Silhouette Score	Higher is Better (↑)	Lower is Better (↓)
Davies-Bouldin Index	Lower is Better (↓)	Higher is Better (↑)
Interpretability	High (Real medoid points)	Medium (Abstract centroids)
Scalability	O(n²) sample-based	O(n²) sample-based



7.3 Winner Selection

The system automatically compares both Silhouette Scores and selects the best-performing model. K-Medoids demonstrated superior cluster separation on the Smart Campus dataset, making it the primary model for the final deployment pipeline.

8. Deployment & Final Insights (Task 7)

8.1 SmartCampusSystem Class

The final system was encapsulated in a SmartCampusSystem class that integrates all components into a single, cohesive inference pipeline. Given a raw sensor reading, the system performs the following steps in sequence:

#	Stage	Description
1	Preprocessing	Apply StandardScaler and select GA-optimized features
2	K-Medoids Prediction	Compute L2 distance to each stored medoid; assign nearest cluster
3	Fuzzy Risk Assessment	Map scaled features to usage/latency inputs; run FIS to get risk score
4	Decision Generation	Map (cluster, risk) pair to an actionable facility management command

8.2 Decision Logic

Each space receives one of three management commands based on its assigned cluster:

- **Cluster 0 → Energy Saving Mode: Low Demand – Reduce HVAC load, dim lighting, pause non-essential devices.**
- Cluster 1 → Optimal Mode: Normal Campus Activity – Maintain current settings, no intervention needed.
- Cluster 2 → High Alert: Critical Resource Usage – Trigger optimization routine or escalate to facility management.

8.3 Key Insights & Recommendations

The following strategic recommendations emerge from the full pipeline analysis:

- **Off-peak automation: Cluster 0 spaces during nighttime hours can have HVAC and lighting reduced by 40-60%, yielding significant cost savings.**
- Proactive maintenance: High-risk Fuzzy scores (>70) in otherwise low-occupancy spaces indicate equipment inefficiency requiring inspection.
- Dynamic scheduling: Temporal patterns in energy consumption support predictive load scheduling aligned with lecture timetables.
- Scalable deployment: The sample-and-extend strategy employed in K-Medoids proves viable for full campus-scale real-time inference.

9. Conclusion

This project successfully designed, implemented, and validated a complete intelligent energy management pipeline for a Smart Campus environment. By integrating four distinct AI/data mining techniques — Genetic Algorithms, K-Medoids Clustering, Hierarchical Clustering, and Fuzzy Logic — into a cohesive end-to-end system, the project demonstrates that data-driven automation can substantially improve energy efficiency in institutional settings.

The pipeline achieved the following key outcomes:

- Efficient dimensionality reduction through GA-based feature selection, improving clustering quality while reducing computational overhead.
- Robust and interpretable spatial grouping via K-Medoids, confirmed by Hierarchical Clustering analysis.
- Nuanced risk quantification through a 9-rule Fuzzy Inference System, handling the inherent ambiguity in energy waste assessment.
- A deployable, scalable SmartCampusSystem class capable of processing live sensor readings and issuing automated facility management commands.

Future extensions could incorporate time-series forecasting (LSTM), reinforcement learning for adaptive HVAC control, and integration with real IoT sensor APIs for live campus deployment.